Using the Formal Definitions

Use the formal definitions of limits as $x \to \pm\infty$ to establish the limits in Exercises 91 and 92.

91. If f has the constant value $f(x) = k$, then $\lim_{x\to\infty} f(x) = k$.

92. If f has the constant value $f(x) = k$, then $\lim_{x\to-\infty} f(x) = k$.

Use formal definitions to prove the limit statements in Exercises 93–96.

93. $\lim_{x\to 0} \dfrac{-1}{x^2} = -\infty$

94. $\lim_{x\to 0} \dfrac{1}{|x|} = \infty$

95. $\lim_{x\to 3} \dfrac{-2}{(x-3)^2} = -\infty$

96. $\lim_{x\to -5} \dfrac{1}{(x+5)^2} = \infty$

97. Here is the definition of an **infinite right-hand limit**.

> Suppose that an interval (c, d) lies in the domain of f. We say that $f(x)$ approaches infinity as x *approaches* c from the right, and write
>
> $$\lim_{x\to c^+} f(x) = \infty,$$
>
> if, for every positive real number B, there exists a corresponding number $\delta > 0$ such that
>
> $$f(x) > B \quad \text{whenever} \quad c < x < c + \delta.$$

Modify the definition to cover the following cases.

a. $\lim_{x\to c^-} f(x) = \infty$

b. $\lim_{x\to c^+} f(x) = -\infty$

c. $\lim_{x\to c^-} f(x) = -\infty$

Use the formal definitions from Exercise 97 to prove the limit statements in Exercises 98–102.

98. $\lim_{x\to 0^+} \dfrac{1}{x} = \infty$

99. $\lim_{x\to 0^-} \dfrac{1}{x} = -\infty$

100. $\lim_{x\to 2^-} \dfrac{1}{x-2} = -\infty$

101. $\lim_{x\to 2^+} \dfrac{1}{x-2} = \infty$

102. $\lim_{x\to 1^-} \dfrac{1}{1-x^2} = \infty$

Oblique Asymptotes

Graph the rational functions in Exercises 103–108. Include the graphs and equations of the asymptotes.

103. $y = \dfrac{x^2}{x-1}$

104. $y = \dfrac{x^2+1}{x-1}$

105. $y = \dfrac{x^2-4}{x-1}$

106. $y = \dfrac{x^2-1}{2x+4}$

107. $y = \dfrac{x^2-1}{x}$

108. $y = \dfrac{x^3+1}{x^2}$

Additional Graphing Exercises

T Graph the curves in Exercises 109–112. Explain the relationship between the curve's formula and what you see.

109. $y = \dfrac{x}{\sqrt{4-x^2}}$

110. $y = \dfrac{-1}{\sqrt{4-x^2}}$

111. $y = x^{2/3} + \dfrac{1}{x^{1/3}}$

112. $y = \sin\left(\dfrac{\pi}{x^2+1}\right)$

T Graph the functions in Exercises 113 and 114. Then answer the following questions.

a. How does the graph behave as $x \to 0^+$?

b. How does the graph behave as $x \to \pm\infty$?

c. How does the graph behave near $x = 1$ and $x = -1$?

Give reasons for your answers.

113. $y = \dfrac{3}{2}\left(x - \dfrac{1}{x}\right)^{2/3}$

114. $y = \dfrac{3}{2}\left(\dfrac{x}{x-1}\right)^{2/3}$

CHAPTER 2 Questions to Guide Your Review

1. What is the average rate of change of the function $g(t)$ over the interval from $t = a$ to $t = b$? How is it related to a secant line?
2. What limit must be calculated to find the rate of change of a function $g(t)$ at $t = t_0$?
3. Give an informal or intuitive definition of the limit
   $$\lim_{x\to c} f(x) = L.$$
   Why is the definition "informal"? Give examples.
4. Does the existence and value of the limit of a function $f(x)$ as x approaches c ever depend on what happens at $x = c$? Explain and give examples.
5. What function behaviors might occur for which the limit may fail to exist? Give examples.
6. What theorems are available for calculating limits? Give examples of how the theorems are used.
7. How are one-sided limits related to limits? How can this relationship sometimes be used to calculate a limit or prove it does not exist? Give examples.
8. What is the value of $\lim_{\theta\to 0} ((\sin\theta)/\theta)$? Does it matter whether θ is measured in degrees or radians? Explain.
9. What exactly does $\lim_{x\to c} f(x) = L$ mean? Give an example in which you find a $\delta > 0$ for a given f, L, c, and $\varepsilon > 0$ in the precise definition of limit.
10. Give precise definitions of the following statements.
    a. $\lim_{x\to 2^-} f(x) = 5$ **b.** $\lim_{x\to 2^+} f(x) = 5$
    c. $\lim_{x\to 2} f(x) = \infty$ **d.** $\lim_{x\to 2} f(x) = -\infty$
11. What conditions must be satisfied by a function if it is to be continuous at an interior point of its domain? At an endpoint?

12. How can looking at the graph of a function help you tell where the function is continuous?

13. What does it mean for a function to be right-continuous at a point? Left-continuous? How are continuity and one-sided continuity related?

14. What does it mean for a function to be continuous on an interval? Give examples to illustrate the fact that a function that is not continuous on its entire domain may still be continuous on selected intervals within the domain.

15. What are the basic types of discontinuity? Give an example of each. What is a removable discontinuity? Give an example.

16. What does it mean for a function to have the Intermediate Value Property? What conditions guarantee that a function has this property over an interval? What are the consequences for graphing and solving the equation $f(x) = 0$?

17. Under what circumstances can you extend a function $f(x)$ to be continuous at a point $x = c$? Give an example.

18. What exactly do $\lim_{x\to\infty} f(x) = L$ and $\lim_{x\to-\infty} f(x) = L$ mean? Give examples.

19. What are $\lim_{x\to\pm\infty} k$ (k a constant) and $\lim_{x\to\pm\infty} (1/x)$? How do you extend these results to other functions? Give examples.

20. How do you find the limit of a rational function as $x \to \pm\infty$? Give examples.

21. What are horizontal and vertical asymptotes? Give examples.

CHAPTER 2 Practice Exercises

Limits and Continuity

1. Graph the function

$$f(x) = \begin{cases} 1, & x \le -1 \\ -x, & -1 < x < 0 \\ 1, & x = 0 \\ -x, & 0 < x < 1 \\ 1, & x \ge 1. \end{cases}$$

Then discuss, in detail, limits, one-sided limits, continuity, and one-sided continuity of f at $x = -1$, 0, and 1. Are any of the discontinuities removable? Explain.

2. Repeat the instructions of Exercise 1 for

$$f(x) = \begin{cases} 0, & x \le -1 \\ 1/x, & 0 < |x| < 1 \\ 0, & x = 1 \\ 1, & x > 1. \end{cases}$$

3. Suppose that $f(t)$ and $f(t)$ are defined for all t and that $\lim_{t\to t_0} f(t) = -7$ and $\lim_{t\to t_0} g(t) = 0$. Find the limit as $t \to t_0$ of the following functions.

a. $3f(t)$ **b.** $(f(t))^2$

c. $f(t)\cdot g(t)$ **d.** $\dfrac{f(t)}{g(t) - 7}$

e. $\cos(g(t))$ **f.** $|f(t)|$

g. $f(t) + g(t)$ **h.** $1/f(t)$

4. Suppose the functions $f(x)$ and $g(x)$ are defined for all x and that $\lim_{x\to0} f(x) = 1/2$ and $\lim_{x\to0} g(x) = \sqrt{2}$. Find the limits as $x \to 0$ of the following functions.

a. $-g(x)$ **b.** $g(x)\cdot f(x)$

c. $f(x) + g(x)$ **d.** $1/f(x)$

e. $x + f(x)$ **f.** $\dfrac{f(x)\cdot\cos x}{x - 1}$

In Exercises 5 and 6, find the value that $\lim_{x\to0} g(x)$ must have if the given limit statements hold.

5. $\lim_{x\to0}\left(\dfrac{4 - g(x)}{x}\right) = 1$

6. $\lim_{x\to-4}\left(x \lim_{x\to0} g(x)\right) = 2$

7. On what intervals are the following functions continuous?

a. $f(x) = x^{1/3}$ **b.** $g(x) = x^{3/4}$

c. $h(x) = x^{-2/3}$ **d.** $k(x) = x^{-1/6}$

8. On what intervals are the following functions continuous?

a. $f(x) = \tan x$

b. $g(x) = \csc x$

c. $h(x) = \dfrac{\cos x}{x - \pi}$

d. $k(x) = \dfrac{\sin x}{x}$

Finding Limits

In Exercises 9–28, find the limit or explain why it does not exist.

9. $\lim \dfrac{x^2 - 4x + 4}{x^3 + 5x^2 - 14x}$

a. as $x \to 0$ **b.** as $x \to 2$

10. $\lim \dfrac{x^2 + x}{x^5 + 2x^4 + x^3}$

a. as $x \to 0$ **b.** as $x \to -1$

11. $\lim_{x\to1} \dfrac{1 - \sqrt{x}}{1 - x}$

12. $\lim_{x\to a} \dfrac{x^2 - a^2}{x^4 - a^4}$

13. $\lim_{h\to0} \dfrac{(x + h)^2 - x^2}{h}$

14. $\lim_{x\to0} \dfrac{(x + h)^2 - x^2}{h}$

15. $\lim_{x\to0} \dfrac{\dfrac{1}{2 + x} - \dfrac{1}{2}}{x}$

16. $\lim_{x\to0} \dfrac{(2 + x)^3 - 8}{x}$

17. $\lim_{x\to1} \dfrac{x^{1/3} - 1}{\sqrt{x} - 1}$

18. $\lim_{x\to64} \dfrac{x^{2/3} - 16}{\sqrt{x} - 8}$

19. $\lim_{x\to0} \dfrac{\tan(2x)}{\tan(\pi x)}$

20. $\lim_{x\to\pi^-} \csc x$

21. $\lim_{x\to\pi} \sin\left(\dfrac{x}{2} + \sin x\right)$

22. $\lim_{x\to\pi} \cos^2(x - \tan x)$

23. $\lim_{x\to0} \dfrac{8x}{3\sin x - x}$

24. $\lim_{x\to0} \dfrac{\cos 2x - 1}{\sin x}$

In Exercises 25–28, find the limit of $g(x)$ as x approaches the indicated value.

25. $\lim_{x\to 0^+} (4g(x))^{1/3} = 2$

26. $\lim_{x\to\sqrt{5}} \dfrac{1}{x + g(x)} = 2$

27. $\lim_{x\to 1} \dfrac{3x^2 + 1}{g(x)} = \infty$

28. $\lim_{x\to -2} \dfrac{5 - x^2}{\sqrt{g(x)}} = 0$

T **Roots**

29. Let $f(x) = x^3 - x - 1$.

a. Use the Intermediate Value Theorem to show that f has a zero between -1 and 2.

b. Solve the equation $f(x) = 0$ graphically with an error of magnitude at most 10^{-8}.

c. It can be shown that the exact value of the solution in part **(b)** is

$$\left(\frac{1}{2} + \frac{\sqrt{69}}{18}\right)^{1/3} + \left(\frac{1}{2} - \frac{\sqrt{69}}{18}\right)^{1/3}.$$

Evaluate this exact answer and compare it with the value you found in part (b).

T **30.** Let $f(\theta) = \theta^3 - 2\theta + 2$.

a. Use the Intermediate Value Theorem to show that f has a zero between -2 and 0.

b. Solve the equation $f(\theta) = 0$ graphically with an error of magnitude at most 10^{-4}.

c. It can be shown that the exact value of the solution in part (b) is

$$\left(\sqrt{\frac{19}{27}} - 1\right)^{1/3} - \left(\sqrt{\frac{19}{27}} + 1\right)^{1/3}.$$

Evaluate this exact answer and compare it with the value you found in part (b).

Continuous Extension

31. Can $f(x) = x(x^2 - 1)/|x^2 - 1|$ be extended to be continuous at $x = 1$ or -1? Give reasons for your answers. (Graph the function—you will find the graph interesting.)

32. Explain why the function $f(x) = \sin(1/x)$ has no continuous extension to $x = 0$.

T In Exercises 33–36, graph the function to see whether it appears to have a continuous extension to the given point a. If it does, use Trace and Zoom to find a good candidate for the extended function's value at a. If the function does not appear to have a continuous extension, can it be extended to be continuous from the right or left? If so, what do you think the extended function's value should be?

33. $f(x) = \dfrac{x - 1}{x - \sqrt[4]{x}}, \quad a = 1$

34. $g(\theta) = \dfrac{5\cos\theta}{4\theta - 2\pi}, \quad a = \pi/2$

35. $h(t) = (1 + |t|)^{1/t}, \quad a = 0$

36. $k(x) = \dfrac{x}{1 - 2^{|x|}}, \quad a = 0$

Limits at Infinity

Find the limits in Exercises 37–46.

37. $\lim_{x\to\infty} \dfrac{2x + 3}{5x + 7}$

38. $\lim_{x\to -\infty} \dfrac{2x^2 + 3}{5x^2 + 7}$

39. $\lim_{x\to -\infty} \dfrac{x^2 - 4x + 8}{3x^3}$

40. $\lim_{x\to\infty} \dfrac{1}{x^2 - 7x + 1}$

41. $\lim_{x\to -\infty} \dfrac{x^2 - 7x}{x + 1}$

42. $\lim_{x\to\infty} \dfrac{x^4 + x^3}{12x^3 + 128}$

43. $\lim_{x\to\infty} \dfrac{\sin x}{\lfloor x \rfloor}$ (If you have a grapher, try graphing the function for $-5 \le x \le 5$.)

44. $\lim_{\theta\to\infty} \dfrac{\cos\theta - 1}{\theta}$ (If you have a grapher, try graphing $f(x) = x(\cos(1/x) - 1)$ near the origin to "see" the limit at infinity.)

45. $\lim_{x\to\infty} \dfrac{x + \sin x + 2\sqrt{x}}{x + \sin x}$

46. $\lim_{x\to\infty} \dfrac{x^{2/3} + x^{-1}}{x^{2/3} + \cos^2 x}$

Horizontal and Vertical Asymptotes

47. Use limits to determine the equations for all vertical asymptotes.

a. $y = \dfrac{x^2 + 4}{x - 3}$

b. $f(x) = \dfrac{x^2 - x - 2}{x^2 - 2x + 1}$

c. $y = \dfrac{x^2 + x - 6}{x^2 + 2x - 8}$

48. Use limits to determine the equations for all horizontal asymptotes.

a. $y = \dfrac{1 - x^2}{x^2 + 1}$

b. $f(x) = \dfrac{\sqrt{x} + 4}{\sqrt{x} + 4}$

c. $g(x) = \dfrac{\sqrt{x^2 + 4}}{x}$

d. $y = \sqrt{\dfrac{x^2 + 9}{9x^2 + 1}}$

49. Determine the domain and range of $y = \dfrac{\sqrt{16 - x^2}}{x - 2}$.

50. Assume that constants a and b are positive. Find equations for all horizontal and vertical asymptotes for the graph of $y = \dfrac{\sqrt{ax^2 + 4}}{x - b}$.

CHAPTER 2 Additional and Advanced Exercises

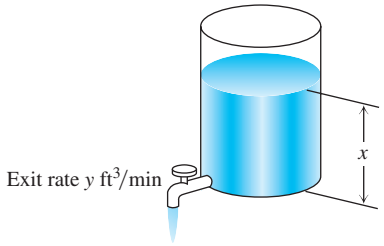
1. Lorentz contraction In relativity theory, the length of an object, say a rocket, appears to an observer to depend on the speed at which the object is traveling with respect to the observer. If the observer measures the rocket's length as L_0 at rest, then at speed v the length will appear to be

$$L = L_0\sqrt{1 - \frac{v^2}{c^2}}.$$

This equation is the Lorentz contraction formula. Here, c is the speed of light in a vacuum, about 3×10^8 m/sec. What happens to L as v increases? Find $\lim_{v\to c^-} L$. Why was the left-hand limit needed?

2. Controlling the flow from a draining tank Torricelli's law says that if you drain a tank like the one in the figure shown, the rate y at which water runs out is a constant times the square root of the

water's depth x. The constant depends on the size and shape of the exit valve.



Suppose that $y = \sqrt{x}/2$ for a certain tank. You are trying to maintain a fairly constant exit rate by adding water to the tank with a hose from time to time. How deep must you keep the water if you want to maintain the exit rate

a. within 0.2 ft^3/min of the rate $y_0 = 1$ ft^3/min?

b. within 0.1 ft^3/min of the rate $y_0 = 1$ ft^3/min?

3. Thermal expansion in precise equipment As you may know, most metals expand when heated and contract when cooled. The dimensions of a piece of laboratory equipment are sometimes so critical that the shop where the equipment is made must be held at the same temperature as the laboratory where the equipment is to be used. A typical aluminum bar that is 10 cm wide at 70°F will be

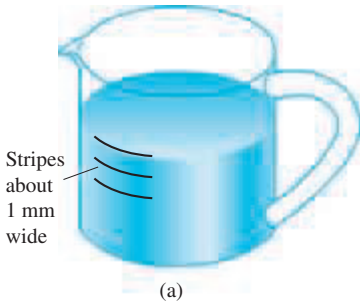
$$y = 10 + (t - 70) \times 10^{-4}$$

centimeters wide at a nearby temperature t. Suppose that you are using a bar like this in a gravity wave detector, where its width must stay within 0.0005 cm of the ideal 10 cm. How close to $t_0 = 70$°F must you maintain the temperature to ensure that this tolerance is not exceeded?

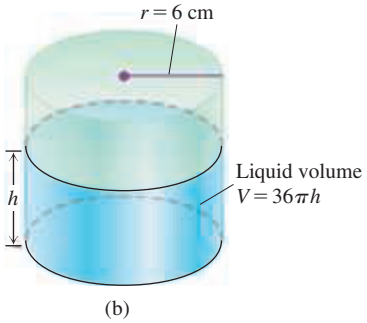
4. Stripes on a measuring cup The interior of a typical 1-L measuring cup is a right circular cylinder of radius 6 cm (see accompanying figure). The volume of water we put in the cup is therefore a function of the level h to which the cup is filled, the formula being

$$V = \pi 6^2 h = 36\pi h.$$

How closely must we measure h to measure out 1 L of water (1000 cm^3) with an error of no more than 1% (10 cm^3)?



(a)



(b)

A 1-L measuring cup (a), modeled as a right circular cylinder (b) of radius $r = 6$ cm

Precise Definition of Limit

In Exercises 5–8, use the formal definition of limit to prove that the function is continuous at c.

5. $f(x) = x^2 - 7, \quad c = 1$ **6.** $g(x) = 1/(2x), \quad c = 1/4$

7. $h(x) = \sqrt{2x - 3}, \quad c = 2$ **8.** $F(x) = \sqrt{9 - x}, \quad c = 5$

9. Uniqueness of limits Show that a function cannot have two different limits at the same point. That is, if $\lim_{x\to c} f(x) = L_1$ and $\lim_{x\to c} f(x) = L_2$, then $L_1 = L_2$.

10. Prove the limit Constant Multiple Rule:

$$\lim_{x\to c} kf(x) = k \lim_{x\to c} f(x) \quad \text{for any constant } k.$$

11. One-sided limits If $\lim_{x\to 0^+} f(x) = A$ and $\lim_{x\to 0^-} f(x) = B$, find

a. $\lim_{x\to 0^+} f(x^3 - x)$ **b.** $\lim_{x\to 0^-} f(x^3 - x)$

c. $\lim_{x\to 0^+} f(x^2 - x^4)$ **d.** $\lim_{x\to 0^-} f(x^2 - x^4)$

12. Limits and continuity Which of the following statements are true, and which are false? If true, say why; if false, give a counterexample (that is, an example confirming the falsehood).

a. If $\lim_{x\to c} f(x)$ exists but $\lim_{x\to c} g(x)$ does not exist, then $\lim_{x\to c}(f(x) + g(x))$ does not exist.

b. If neither $\lim_{x\to c} f(x)$ nor $\lim_{x\to c} g(x)$ exists, then $\lim_{x\to c}(f(x) + g(x))$ does not exist.

c. If f is continuous at x, then so is $|f|$.

d. If $|f|$ is continuous at c, then so is f.

In Exercises 13 and 14, use the formal definition of limit to prove that the function has a continuous extension to the given value of x.

13. $f(x) = \dfrac{x^2 - 1}{x + 1}, \quad x = -1$ **14.** $g(x) = \dfrac{x^2 - 2x - 3}{2x - 6}, \quad x = 3$

15. A function continuous at only one point Let

$$f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

a. Show that f is continuous at $x = 0$.

b. Use the fact that every nonempty open interval of real numbers contains both rational and irrational numbers to show that f is not continuous at any nonzero value of x.

16. The Dirichlet ruler function If x is a rational number, then x can be written in a unique way as a quotient of integers m/n where $n > 0$ and m and n have no common factors greater than 1. (We say that such a fraction is in *lowest terms*. For example, $6/4$ written in lowest terms is $3/2$.) Let $f(x)$ be defined for all x in the interval $[0, 1]$ by

$$f(x) = \begin{cases} 1/n, & \text{if } x = m/n \text{ is a rational number in lowest terms} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

For instance, $f(0) = f(1) = 1$, $f(1/2) = 1/2$, $f(1/3) = f(2/3) = 1/3$, $f(1/4) = f(3/4) = 1/4$, and so on.

a. Show that f is discontinuous at every rational number in $[0, 1]$.

b. Show that f is continuous at every irrational number in $[0, 1]$. (*Hint*: If ε is a given positive number, show that there are only finitely many rational numbers r in $[0, 1]$ such that $f(r) \geq \varepsilon$.)

c. Sketch the graph of f. Why do you think f is called the "ruler function"?

17. Antipodal points Is there any reason to believe that there is always a pair of antipodal (diametrically opposite) points on Earth's equator where the temperatures are the same? Explain.

18. If $\lim_{x\to c}(f(x) + g(x)) = 3$ and $\lim_{x\to c}(f(x) - g(x)) = -1$, find $\lim_{x\to c} f(x)g(x)$.

19. Roots of a quadratic equation that is almost linear The equation $ax^2 + 2x - 1 = 0$, where a is a constant, has two roots if $a > -1$ and $a \neq 0$, one positive and one negative:

$$r_+(a) = \frac{-1 + \sqrt{1 + a}}{a}, \qquad r_-(a) = \frac{-1 - \sqrt{1 + a}}{a},$$

a. What happens to $r_+(a)$ as $a \to 0$? As $a \to -1^+$?

b. What happens to $r_-(a)$ as $a \to 0$? As $a \to -1^+$?

c. Support your conclusions by graphing $r_+(a)$ and $r_-(a)$ as functions of a. Describe what you see.

d. For added support, graph $f(x) = ax^2 + 2x - 1$ simultaneously for $a = 1, 0.5, 0.2, 0.1$, and 0.05.

20. Root of an equation Show that the equation $x + 2\cos x = 0$ has at least one solution.

21. Bounded functions A real-valued function f is **bounded from above** on a set D if there exists a number N such that $f(x) \leq N$ for all x in D. We call N, when it exists, an **upper bound** for f on D and say that f is bounded from above by N. In a similar manner, we say that f is **bounded from below** on D if there exists a number M such that $f(x) \geq M$ for all x in D. We call M, when it exists, a **lower bound** for f on D and say that f is bounded from below by M. We say that f is **bounded** on D if it is bounded from both above and below.

a. Show that f is bounded on D if and only if there exists a number B such that $|f(x)| \leq B$ for all x in D.

b. Suppose that f is bounded from above by N. Show that if $\lim_{x\to c} f(x) = L$, then $L \leq N$.

c. Suppose that f is bounded from below by M. Show that if $\lim_{x\to c} f(x) = L$, then $L \geq M$.

22. Max $\{a, b\}$ and min $\{a, b\}$

a. Show that the expression

$$\max\{a, b\} = \frac{a + b}{2} + \frac{|a - b|}{2}$$

equals a if $a \geq b$ and equals b if $b \geq a$. In other words, max $\{a, b\}$ gives the larger of the two numbers a and b.

b. Find a similar expression for min $\{a, b\}$, the smaller of a and b.

Generalized Limits Involving $\frac{\sin\theta}{\theta}$

The formula $\lim_{\theta\to 0}(\sin\theta)/\theta = 1$ can be generalized. If $\lim_{x\to c} f(x) = 0$ and $f(x)$ is never zero in an open interval containing the point $x = c$, except possibly at c itself, then

$$\lim_{x\to c}\frac{\sin f(x)}{f(x)} = 1.$$

Here are several examples.

a. $\lim_{x\to 0}\dfrac{\sin x^2}{x^2} = 1$

b. $\lim_{x\to 0}\dfrac{\sin x^2}{x} = \lim_{x\to 0}\dfrac{\sin x^2}{x^2}\lim_{x\to 0}\dfrac{x^2}{x} = 1\cdot 0 = 0$

c. $\lim_{x\to -1}\dfrac{\sin(x^2 - x - 2)}{x + 1} = \lim_{x\to -1}\dfrac{\sin(x^2 - x - 2)}{(x^2 - x - 2)}.$

$$\lim_{x\to -1}\frac{(x^2 - x - 2)}{x + 1} = 1\cdot\lim_{x\to -1}\frac{(x + 1)(x - 2)}{x + 1} = -3$$

d. $\lim_{x\to 1}\dfrac{\sin\left(1 - \sqrt{x}\right)}{x - 1} = \lim_{x\to 1}\dfrac{\sin\left(1 - \sqrt{x}\right)}{1 - \sqrt{x}}\dfrac{1 - \sqrt{x}}{x - 1}$

$$= \lim_{x\to 1}\frac{\left(1 - \sqrt{x}\right)\left(1 + \sqrt{x}\right)}{(x - 1)\left(1 + \sqrt{x}\right)}$$

$$= \lim_{x\to 1}\frac{1 - x}{(x - 1)\left(1 + \sqrt{x}\right)} = -\frac{1}{2}$$

Find the limits in Exercises 23–28.

23. $\lim_{x\to 0}\dfrac{\sin(1 - \cos x)}{x}$

24. $\lim_{x\to 0^+}\dfrac{\sin x}{\sin\sqrt{x}}$

25. $\lim_{x\to 0}\dfrac{\sin(\sin x)}{x}$

26. $\lim_{x\to 0}\dfrac{\sin(x^2 + x)}{x}$

27. $\lim_{x\to 2}\dfrac{\sin(x^2 - 4)}{x - 2}$

28. $\lim_{x\to 9}\dfrac{\sin\left(\sqrt{x} - 3\right)}{x - 9}$

Oblique Asymptotes

Find all possible oblique asymptotes in Exercises 29–32.

29. $y = \dfrac{2x^{3/2} + 2x - 3}{\sqrt{x} + 1}$

30. $y = x + x\sin\dfrac{1}{x}$

31. $y = \sqrt{x^2 + 1}$

32. $y = \sqrt{x^2 + 2x}$

Showing an Equation Is Solvable

33. Assume that $1 < a < b$ and $\dfrac{a}{x} + x = \dfrac{1}{x - b}$. Show that this equation is solvable.

More Limits

34. Find constants a and b so that each of the following limits is true.

a. $\lim_{x\to 0} \dfrac{\sqrt{a+bx}-1}{x} = 2$ **b.** $\lim_{x\to 1} \dfrac{\tan(ax-a)+b-2}{x-1} = 3$

35. Evaluate $\lim_{x\to 1} \dfrac{x^{2/3}-1}{1-\sqrt{x}}$. **36.** Evaluate $\lim_{x\to 0} \dfrac{|3x+4|-|x|-4}{x}$.

Limits on Arbitrary Domains

The definition of the limit of a function at $x = c$ extends to functions whose domains near c are more complicated than intervals.

> **General Definition of Limit**
> Suppose every open interval containing c contains a point other than c in the domain of f. We say that $\lim_{x\to c} f(x) = L$ if for every number $\varepsilon > 0$ there exists a corresponding number $\delta > 0$ such that for all x in the domain of f, $|f(x) - L| < \varepsilon$ whenever $0 < |x - c| < \delta$.

For the functions in Exercises 37–39,

a. Find the domain.

b. Show that at $c = 0$ the domain has the property described above.

c. Evaluate $\lim_{x\to 0} f(x)$.

37. The function f is defined as follows: $f(x) = x$ if $x = 1/n$ where n is a positive integer, and $f(0) = 1$.

38. The function f is defined as follows: $f(x) = 1 - x$ if $x = 1/n$ where n is a positive integer, and $f(0) = 1$.

39. $f(x) = \sqrt{x \sin(1/x)}$

40. Let g be a function with domain the rational numbers, defined by $g(x) = \dfrac{2}{x-\sqrt{2}}$ for rational x.

a. Sketch the graph of g as well as you can, keeping in mind that g is only defined at rational points.

b. Use the general definition of a limit to prove that $\lim_{x\to 0} g(x) = -\sqrt{2}$.

c. Prove that g is continuous at the point $x = 0$ by showing that the limit in part (b) equals $g(0)$.

d. Is g continuous at other points of its domain?

CHAPTER 2 Technology Application Projects

Mathematica/Maple Projects

Projects can be found within MyMathLab.

Take It to the Limit
Part I
Part II (Zero Raised to the Power Zero: What Does It Mean?)
Part III (One-Sided Limits)
Visualize and interpret the limit concept through graphical and numerical explorations.
Part IV (What a Difference a Power Makes)
See how sensitive limits can be with various powers of x.

Going to Infinity
Part I (Exploring Function Behavior as $x \to \infty$ or $x \to -\infty$)
This module provides four examples to explore the behavior of a function as $x \to \infty$ or $x \to -\infty$.
Part II (Rates of Growth)
Observe graphs that *appear* to be continuous, yet the function is not continuous. Several issues of continuity are explored to obtain results that you may find surprising.